MILITARY SPECIFICATION

OUTBOARD MOTOR, GASOLINE: 18-20 HP, WITH CARRYING CHEST

1. SCOPE

* 1.1 This specification covers the requirements for one type of outboard motor powered by an 18- to 20-horsepower (hp) gasoline engine, including maintenance tools, repair parts, and plywood carrying chest.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	- Box, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	-Preservation, Methods of.
MIL-V-173	-Varnish, Moisture-and Fungus-Resistant (For the Treatment of Communication Electronic, and Associated Electrical Equipment).
MIL-R-196	-Repair Parts for Internal Combustion Engines, Packaging of.
MIL-P-514	-Plates; Identification, Instruction and Marking, Blank.
MIL-T-704	-Treatment and Painting of Materiel.
MIL-L-2104	-Lubricating Oil, Internal-Combustion Engine Heavy Duty).
MIL-L-2105	-Lubricating Oil, Gear Multipurpose.
MIL-G-3056	-Gasoline, Automotive, Combat.
MIL-F-3541	-Fittings, Lubrication.
MIL-C-3702	-Cable, Power, Electrical: Ignition, High-Tension.
MIL-G-3859	-Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14-ounce and 21-ounce Capacity).

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MIL-E-10062	-Engines: Preparation for Shipment and Storage of.
MIL-G-10924	-Grease, Automotive and Artillery.
MIL-C-12044	-Chests, Plywood.
MIL-C-13909	-Conduit, Metal, Flexible; Electrical, Shielded.
MIL-H-15424	-Hand Tools, Packaging of.
MIL-I-16165	-Interference Shielding, Engine Electrical Systems.
MIL-G-23827	-Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-130	-Identification Marking of U.S. Military Property.
MIL-STD-461	-Electromagnetic Interference Characteristics Requirements for Equipment.
MS35909	-Spark Plug, Shielded, 14MM, 1-1/4-inch Well (Other than Aircraft).

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28--Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

OUTBOARD BOATING CLUB OF AMERICA

Engineering Manual of Recommended Practices

(Application for copies should be addressed to Outboard Boating Club of America, 307 N. Michigan Avenue, Chicago, Illinois 60601.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue, New York, New York 10016.)

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AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street N.W., Washington, D. C. 20036.)

3. REQUIREMENTS

* 3.1 First article. The first article is defined as a production model identical to the product that the contractor will subsequently deliver in fulfillment of the contract. As soon as practicable after award of the contract or on a date specified in the contract, and prior to submission of any units for final acceptance, the contractor shall furnish a first article for examination and test to determine conformance to this specification. The first article shall consist of the outboard motor, accessories, carrying chest, tools and repair parts. Examination and testing shall be as specified herein. Approval of the first article by the procuring activity shall not relieve the contractor of his obligation to supply a product which conforms to this specification. Any changes or deviations of production units from the first article shall be subject to the approval of the contracting officer.

* 3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor, shall be adequate for the intended use, and shall be subject to all provisions of this specification. Corrosion-resistant materials such as corrosion-resistant steel, copper, brass, bronze, copper-nickel alloys, high-strength aluminum alloys, and reinforced fiber glass shall be used to the maximum practicable extent in the manufacture of the outboard motor.

* 3.3 Design and construction. Each outboard motor shall consist of a gasoline engine, mounting bracket, forward and reverse reduction gears, propeller, and separate fuel tank with connecting fuel lines. The motor leg shall be designed for boats with a 20-inch transom. The motor leg shall be designed to effect minimum drag and to prevent undesirable hydraulic effects, such as water entering the boat over the transom.

* 3.3.1 Engine. The net brake hp of the engine, complete with all accessories, shall be between 18 and 20, continuous, and at the manufacturer's maximum recommended engine speed. The hp rating of the engine shall be determined in accordance with the procedures outlined in the Engineering Manual of Recommended Practices of the Outboard Boating Club of America.

* 3.3.2 Throttle control. The engine throttle control shall be located in the twistgrip on the steering handle and shall synchronize throttle and spark in correct relationship for optimum performance throughout the operating range of the engine. The throttle control shall allow full engine speed for both forward and reverse operations. The throttle control shall operate through a complete cycle of operating limits 2,000 times.

* 3.3.3 Fuel system. The fuel system shall consist of the following:

* 3.3.3.1 Fuel tank. The fuel tank shall be 6 to 7 gallons capacity, shall have a fuel gage, and shall be equipped with either a manual discharge pump or a squeeze bulb. The filler-opening cap shall be attached to the tank by a chain. The fuel tank shall have a carrying handle. The fuel tank shall be fabricated from corrosion-resistant material, or shall be treated and painted to be corrosion-resistant.

* 3.3.3.2 Fuel lines. Fuel lines shall be installed to prevent chafing by moving parts or by vibration, shall be strain-free, and shall contain no loops or rises which will cause vapor lock. A ceramic or sintered bronze fuel filter shall be incorporated in the fuel supply line. A valve shall be provided to close off the fuel line from the fuel tank when the line is disconnected from the outboard motor. The fuel tank fuel line shall be not less than 6 feet long and shall have quick-disconnect fittings for connecting to the engine fuel inlet. The fuel line shall be gasoline and oil resistant and shall not deteriorate due to aging, sunlight, heat, cold, or salt water.

* 3.3.3.3 Carburetor. The carburetor shall be of the venturi type with fixed or adjustable low-speed and high-speed jets. The carburetor shall be equipped with either a manual or automatic choke.

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* 3.3.4 Ignition system. A high-tension magneto ignition system shall be furnished. The system shall include a magneto with waterproof coils, cables, ignition points, spark plugs, and other items necessary for engine starting and operation. Ignition cable shall conform to MIL-C-3702, type I, class 1, grade and size to suit application, and shall be enclosed in a flexible metallic conduit conforming to MIL-C-13909, type 1, grade A. All wiring shall be installed to prevent chafing by moving parts and vibration. Wiring shall not be under strain. The magneto shall be protected from contamination. The magneto shall provide sufficient output to fire the spark plugs at cranking and operating speeds. The ignition points shall be of corrosion-resistant material and shall be replaceable and adjustable without the use of special tools. Ignition timing marks shall be provided. Spark plugs shall conform to MS35909.

* 3.3.5 Cranking system. Engine cranking or starting shall be accomplished by an automatic rewind starter rope. The rope shall be a heavy-duty commercial type. Rope guides shall be installed to prevent chafing. An auxiliary starting sheave shall be incorporated for use in the event the automatic rewind starter becomes inoperative. The automatic-rewind cranking system shall withstand 1,000 cranking operations.

* 3.3.6 Cooling system. The outboard motor shall contain a positive circulating water cooling system. The system shall be self-draining when the outboard motor is in the vertical (operating) position and not running.

* 3.3.7 Exhaust system. An efficient underwater exhaust system shall be provided. The exhaust system shall include an automatic exhaust relief when the outboard motor is running in reverse gear. The automatic exhaust relief shall partially divert exhaust gas from the underwater exit to atmospheric exit to prevent excessive back pressure and propeller cavitation. The exhaust relief system shall become operable automatically when the outboard motor is run in reverse gear.

* 3.3.8 Steering. Steering shall be through an intergral tiller-type handle which shall change direction of the propeller thrust. Steering operation shall be through a total arc of 120° minimum; 60° to right and 60° to left of center. The steering handle shall be hinged for compactness. Operation of the steering handle through full operational arc (extreme right to extreme left) shall not affect or interfere with the speed control operation.

* 3.3.9 Gear shift. The gear shift lever shall be located on the steering handle. The gear shifting system shall operate the power transmission system through a complete shifting cycle 2,000 times.

* 3.3.10 Power transmission. Power to the propeller shall be transmitted from a vertical drive shaft through heavy-duty reduction gears to a horizontal propeller shaft. The gear case shall be watertight.

* 3.3.11 Propeller. The propeller shall be designed to deliver maximum thrust from available power without creating cavitation or other undesirable hydraulic effects. The propeller shall have not less than three blades. The propeller shall be fabricated of manganese bronze or heat-treated, semipermanent or permanent mold, aluminum-alloy casting. Propeller size (diameter, blade area, and pitch) shall be for heavy-duty operation. A replaceable shear pin of the proper shear strength or a slip-type clutch of equal effectiveness shall be provided for protection of the propeller.

* 3.3.12 Fuel mixture. The outboard motor shall be tested and shall operate as specified herein on a fuel-oil mixture of gasoline and lubricating oil; proportions shall be recommended by the manufacturer. The fuel shall be gasoline conforming to MIL-G-3056, type I; the lubricating oil shall conform to MIL-L-2104, grade as recommended by the manufacturer (see 6.3).

* 3.3.13 Lubrication. The outboard motor shall operate as specified herein when Military lubricants of the proper type for the intended purpose are used (see 6.3).

* 3.3.13.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

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* 3.3.13.2 Pressure-release device. A suitable pressure-release device shall be provided when the use of 1,000 pounds-force per square inch, and higher, pressure lubricating equipment will damage grease seals or other parts.

* 3.3.13.3 Lubricant reservoirs. Enclosures, such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks, finger-holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant. Each enclosure shall be equipped with a means for draining the lubricant from the reservoir, located so that removal of the drain plug will result in complete drainage of the enclosure. Drainage shall be to the ground when the outboard motor is in normal operating position (vertical). Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drain plug, the filling means, and the lubricant level check shall be possible without the removal or adjustment of accessories or parts. Engine housing panels equipped with hand-operable, quick-disconnect fasteners may be removed.

* 3.3.14 Mounting bracket. The outboard motor mounting bracket shall provide secure attachment to transoms and motor brackets of boats. The bracket shall provide adjustable slopes from 0° to 20° (minimum) from the vertical without impairment of steering or free tilting in forward and neutral operation. The bracket shall support and sustain the outboard motor while operating at full power in forward and reverse gears, and shall incorporate provisions to minimize vibration. The outboard motor shall not tilt while operating in reverse at all speeds.

* 3.3.15 Housing. Outboard motor components, such as engine, gears, pumps, and fly-wheel shall be enclosed in housings of corrosion-resistant material. The housings shall permit access, without the use of tools, to spark plugs and carburetor for service and repair. The housings shall protect the components from damage and shall afford protection to operating personnel from the hazards of moving parts, high temperatures, and electric shock.

* 3.3.16 Lifting and carrying. The outboard motor shall be provided with a lifting and carrying handle. The handle shall permit the outboard motor to be manually lifted to the normal operating position, and shall permit hand carrying in an approximately horizontal position. The handle shall permit the outboard motor to be carried and positioned when the engine is hot due to recent operation.

* 3.3.17 Safety attachment. Means shall be provided for attaching a 1/2-inch-diameter safety line to the outboard motor.

* 3.3.18 Chest. Each outboard motor shall be furnished with a chest. The chest shall accommodate the completely assembled outboard motor, the fuel tank, fuel lines, maintenance tools, repair parts, and required technical publications. The chest shall be constructed in conformance to MIL-C-12044 and shall not be damaged by free fall test specified in 4.6.12. The contents of the chest shall be arranged in as compact a manner as practicable. All contents of the chest shall be blocked and securely fastened to prevent shifting or damage to the outboard motor or other contents during shipping or field usage. The outboard motor shall be securely fastened in a horizontal position. The fastening device shall be removable with conventional hand tools. The contents of the chest shall remain secure when tested as specified in 4.6.12.

* 3.3.19 Maintenance tools. One set of maintenance tools (including ignition point file) normally furnished as standard equipment, and sufficient to make emergency repairs, shall be furnished in the chest with each outboard motor.

* 3.3.20 Repair parts. The following repair parts shall be furnished in the chest with each outboard motor:

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ITEM	QUANTITY
Starter rope	1
Cotter pin, propeller nut	4
Nut, propeller	1
Shear pin (when required)	6
Propeller	1
Spark Plugs	4 sets
Ignition points assembly with condenser	2
Impeller, water pump	1

* 3.3.21 Technical publications. The contractor shall prepare an operation and maintenance manual. Two copies of the manual shall be supplied with each outboard motor. The manual shall include, but not be limited to, the following information: complete operational and maintenance instructions, a complete listing of all parts incorporated in the outboard motor, a list of repair parts and maintenance tools supplied with the outboard motor, and a listing of operating supplies required for a period of 10-hour operation. The manual shall be applicable to cover the specific equipment furnished. The contractor shall furnish the manual to the contracting officer for review and approval.

* 3.3.22 Fungus and water resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions :

(a) Components or circuit elements which are inherently fungus-and water-resistant or which are hermetically sealed, need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

The varnish shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 0.001 inch to component or element surfaces previously cleaned and prepared in accordance with accepted commercial practices.

* 3.4 Performance. The outboard motor shall operate at full throttle for 75 hours in forward gear, at full throttle for 25 hours in reverse gear, and for an additional 100 hours under cyclic load. The outboard motor shall start and operate as specified herein at ambient temperatures of 45° to 90°F. The 200 hours of operation shall be continuous without maintenance or repair, except that two exhaust port cleanings are permitted and spark plugs may be cleaned or replaced as required. The outboard motor shall operate as specified herein using the fuel-oil mixture specified in 3.3.12. The outboard motor shall deliver the rated hp when operating at the manufacturer's recommended engine speed.

* 3.5 Radio interference suppression. The outboard motor shall be equipped for the reduction of radio interference in conformance to MIL-I-16165 and tested as specified in 4.6.11.

* 3.6 Weight. The maximum unit weight of the outboard motor, not including fuel tank, shall be 85 pounds.

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* 3.7 Treatment and painting. All internal surfaces of the outboard motor and accessories normally painted, and all visible external surfaces of the outboard motor and fuel tank, shall be treated and painted in accordance with MIL-T-704, type B. Color shall match Marine Corps green, no. 23, semigloss (see 6.4). Exceptions shall be electrical equipment (see 3.3.22) and parts subject to high temperature which shall be treated and painted in accordance with standard practice of the manufacturer.

* 3.8 Markings. The markings shall be as follows.

* 3.8.1 Identification. The outboard motor and accessory equipment shall be identified in accordance with MIL-STD-130. Identification markings shall be applied by use of identification plates conforming to MIL-P-514, type IV.

* 3.8.2 Instruction. Each outboard motor shall be equipped with instruction, warning, and caution plates describing any special or important procedures to be followed in starting, operating, and servicing the outboard motor. Instruction plates shall be of corrosion-resistant metal or a durable transfer. Instructions may be included on identification plates (3.8.1) if information can be conveniently included.

* 3.9 Workmanship. The outboard motor shall be furnished clean and free from burrs, rough or sharp edges, dents, and chipped paint. Bolt holes shall be accurately punched or drilled. Washers or lockwashers shall be provided and all fasteners shall be tight. Screw threads shall conform to H28. The outboard motor and accessories shall reflect the highest standards of manufacture consistent with commercial practices within the industry.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Quality conformance (see 4.4).

(c) Preparation for delivery (see 4.7).

* 4.3 First article inspection. The first article outboard motor, selected in accordance with 3.1, shall be subjected to the first article examination and tests specified in Table I.

* 4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following inspections:

(a) Individual. Each outboard motor manufactured for delivery in fulfillment of the contract or order shall be tested as specified in Table I.

(b) Sample. A sample inspection consisting of examination and tests shall be made from each lot of outboard motors manufactured. For purposes of quality conformance inspection, a lot shall consist of the quantity of outboard motors, manufactured at one time or under one continuous production line set-up, to be delivered at one time for fulfillment of one contract or order. Outboard motor shall be selected for sample inspection in accordance with 4.4.1 and 4.4.2.

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* 4.4.1 Sampling for examination. Sampling for examination shall conform to MIL-STD-105, inspection level S-4. Samples selected for examination shall be examined for the defects listed in 4.5. The acceptable quality level (AQL) shall be 1.0 percent defects per hundred units (DHU). Any sample containing one or more defects shall be considered defective.

* 4.4.2 Sampling for tests. Sampling for tests shall conform to MIL-STD-105, inspection level S-1. Samples selected for tests shall be tested as specified in Table I. The AQL shall be 10.0 percent DHU . Any sample containing one or more defects shall be considered defective.

TABLE I. EXAMINATION AND TESTS

Type of Inspection			Examination or Test	Method	Requirement
First Article	Quality Conformance: Individual	Quality Conformance: Sample			
X		X	Examination	4.5	Section 3
		X	Sample Operation	4.6.2	3.3.3.3 3.3.4
	X		Individual Operation	4.6.3	3.3.6 3.3.7
X			Full Throttle Endurance	4.6.4	3.4
X			Cyclic Load Endurance	4.6.5	3.4
X			Throttle Control	4.6.6	3.3.2
X			Cranking	4.6.7	3.3.5
X			Gear Shift	4.6.8	3.3.9
X			Test Run	4.6.9	3.3.3.3 3.3.4 3.3.6 3.3.7 3.3.8 3.3.10 3.3.11 3.3.14 3.3.15
X			Beaching	4.6.10	3.3.8 3.3.10 3.3.14
X			Radio Interference Reduction	4.6.11	3.5
X			Chest	4.6.12	3.3.18
X			Preparation for Delivery	4.7	Section 5

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* 4.5 Examination. The outboard motors, accessories, and chests selected for examination shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

(a) Weight not as required

(b) Material not as required

(c) Fuel tank and fuel tank treatmeant not as required

(d) Fuel lines not as required

(e) Wiring not as required

(f) Housings of improper fit or not as required

(g) Carrying handle not as required

(h) Safety device not as required

(i) Identification and instructions missing or not as required

(j) Painting or treatment not as required

(k) Tools or repair parts incomplete, missing, or not as required

(l) Technical publications missing or not as required

(m) Chest internal blocking and fastening not as required

(n) Workmanship not as required

* 4.6 Tests.

* 4.6.2 Sample operation. The outboard motor shall be operated in forward gear at full throttle, with the propeller in water for a period of 6 hours. Failure of the motor to start, erratic or uneven operation due to improper carburetion or any other reason, failure of water to circulate through the system or overheating for any other reason, improper operation of the exhaust system, evidence of loose or broken parts, or failure or deformation of structural components shall constitute failure of this test.

* 4.6.3 Individual operation. The outboard motor shall be tested as specified in 4.6.2 except that running time shall be for 10 minutes instead of 6 hours.

* 4.6.4 Full throttle endurance. The outboard motor shall be operated in water at full throttle (after any required warm-up period) delivering maximum thrust for a total of 100 hours (75 hours in forward gear and 25 hours in reverse gear). The ambient temperature shall be 45° to 90°F. The operation of the outboard motor shall be observed for conformance to the following operational qualities:

(a) Starting shall be accomplished by not more than five engine cranks.

(b) Engine shall be smooth running with no indication of carburetor or spark maladjustment.

(c) There shall be no excessive vibration.

(d) There shall be no loosening of fittings, parts, or fasteners.

(e) There shall be no overheating due to improper operation of the cooling system.

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(f) The exhaust system shall perform in forward and reverse gear as required in 3.3.7.

(g) Fuel lines and fuel feed system shall supply ample and continuous fuel to engine.

(h) Engine shall run at recommended speeds without adjustment.

(i) Lubrication system for power transmission shall perform properly so that gears and other moving parts do not receive undue or excessive wear evidenced by noise, binding, or excessive backlash.

(j) There shall be no erratic or uneven operation.

(k) There shall be no failure or deformation of any parts.

* 4.6.5 Cycle load endurance. The outboard motor shall be operated in water and in forward gear. The following 10-hour cyclic load schedule shall be performed through 10 full cycles for a total of 100 hours operation. Operation of the outboard motor shall be observed for conformance to the operational qualities specified in 4.6.4, a through k.

Throttle Setting	Duration (Hours)
1/4	1
1/2	1
3/4	2
Full	2
3/4	2
1/2	1
1/4	1

* 4.6.6 Throttle control. The throttle control system shall be operated through a complete cycle of operating limits (minimum to maximum to minimum) 2,000 times. There shall be no deformation, binding, loosening, or failure of any components of the system either during or after the test. Engine shall not be running during this test.

* 4.6.7 Cranking system. The automatic rewind cranking system of the outboard motor under test shall be operated 1,000 times. The fuel line shall be disconnected and the spark plugs shall be installed in the engine during the test. Failure of any component of the cranking system either during or after the test shall constitute failure of the test. Upon completion of the test, the fuel line shall be reconnected and the engine shall start within normal cranking limits.

* 4.6.8 Gear shifting. The gear shifting system of the outboard motor under test shall be operated through 2,000 complete cycles (neutral to forward to neutral to reverse to neutral). The outboard motor shall be operating at maximum shift engine speed during the test. There shall be no evidence of binding, loose components, deformation, or failure during or after the test.

* 4.6.9 Test run. The outboard motor shall be mounted on a small boat (15- to 16-foot) in the water. The boat shall be lightly loaded (2 or 3 passengers) and shall be given a test run. Throttle operation, steering, and gear shifting system shall be checked by turning the boat to port and starboard while running ahead and astern at various speeds up to full throttle. The boat shall be operated and maneuvered ahead at full throttle for not less than 30 minutes. Ambient temperature shall be 45° to 90°F. The operation and handling of the outboard motor shall be observed for conformance to the following operational qualities:

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(a) All requirements of 4.6.4, a through k.

(b) Handling and steering shall operate as specified in 3.3.8.

(c) Gear shift system shall operate at all recommended speeds.

(d) Propeller shall operate as specified in 3.3.11.

(e) Mounting bracket shall not become loose, deformed, or transmit excessive vibration.

(f) Workmanship as specified in 3.9.

Failure of any component shall constitute failure of this test.

* 4.6.10 Beaching. The outboard motor shall be mounted as specified in 4.6.9, except only the operator shall be aboard. The outboard motor shall be run at full throttle and shall propel the boat onto a gently sloping beach. The slope of the beach shall be such that the leg of the outboard motor strikes bottom before the boat runs aground. This test shall be repeated 25 times. Any structural damage to the leg, mounting bracket, or steering system of the outboard motor (excluding the propeller) or failure of the engine to start at any time during the 25 runs shall constitute failure of this test.

* 4.6.11 Radio interference. The outboard motor, shall be tested for radio interference reduction as specified in MIL-STD-461.

* 4.6.12 Chest. The chest to be tested shall be loaded with the outboard motor and all accessories. Positioned horizontally on a hard, solid floor surface, one end of the chest shall be raised and supported on a timber 5 to 6 inches thick. The opposite end of the chest shall be raised to a height of 30 inches and allowed to fall freely. This test shall be repeated five times. The test shall then be conducted with the chest turned on one side, the free-fall height shall be 18 inches, and the test repeated three times. For all tests, the free-fall end shall be the end which contains the power head (engine) of the outboard motor. Damage to or shifting of any item in the chest, loosening or failure of clamping devices, supports or blocking, or damage to the chest other than local compression failure of the member striking the hard surface shall constitute failure of the test.

* 4.7 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in Section 5. The defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in conformance to MIL-STD-105. The lot size shall be the number of shipping containers to be shipped at one time, to one destination, on one contract or order. The inspection level shall be S-1, and the AQL shall be 1.0 DHU.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps or loose strapping.
	Distortion of container.

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5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified in the contract or order (see 6.2).

* 5.1.1 Level A.

* 5.1.1.1 Lubrication for delivery. All components which require lubrication for protection or operation shall be lubricated. Grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. Oil lubrication shall be with oils conforming to MIL-L-2104 or MIL-L-2105, as applicable. The outboard motor shall be serviced with winter grades of lubricants designated for use in temperature above +32°F. Each component shall be tagged in a conspicuous place indicating the temperature range and grade of oils and greases used.

* 5.1.1.2 Engine. The engine shall be preserved in conformance with MIL-E-10062, type I method 1.

* 5.1.1.3 Automatic rewind starter. Metal surfaces of the starter sheave, cable, and spring requiring a preservative in conformance with MIL-P-116 shall be coated with type P-6 preservative.

* 5.1.1.4 Gear case. The gear case shall be filled with lubricant conforming to MIL-L-2105, grade as recommended by the manufacturer, and the gears shall be actuated through a minimum of three complete revolutions to insure coating the gears with the lubricant.

* 5.1.1.5 Technical publications. Technical publications for each outboard motor shall be preserved in conformance to MIL-P-116, method IC-3.

* 5.1.1.6 Repair parts. Repair parts shall be preserved and packaged as specified in MIL-R-196 and in conformance to MIL-P-116. Repair parts shall be placed in the chest.

* 5.1.1.7 Maintenance tools. Maintenance tools shall be preserved and packaged in conformance to MIL-H-15424. Maintenance tools shall be placed in the chest.

* 5.1.1.8 Intermediate packaging. Each outboard motor and all accessories comprising a complete unit shall be intermediate packaged in the chest.

* 5.1.1.9 Shipment. No shipment shall be made unless repair parts, maintenance tools, and accessories are included with the shipment, unless approval for shipment of the outboard motor without such items has been received from the contracting officer.

* 5.1.2 Level C. Each outboard motor and all accessories comprising a complete unit shall be preserved and packaged in a manner to afford protection against corrosion, deterioration, and damage and placed in the chest.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Each chest containing a complete outboard motor unit shall be packed in a box conforming to PPP-B-621, class 2, style optional for type I load. The box shall be modified by the installation of skids to conform to the appendix of the box specification.

* 5.2.2 Level B. Each chest containing a complete outboard motor unit shall be packed in a box conforming to PPP-B-621, class 1, style optional for type 1 load.

* 5.2.3 Level C. Each chest containing a complete outboard motor unit shall be packed in a container to insure carrier acceptance and safe arrival at destination at lowest rates, in conformance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or with other carrier rules applicable to the mode of transportation.

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* 5.3 Marking. In addition to markings required by the contract or order, packages, chest, and shipping container shall be marked in conformance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The outboard motor is intended for use in propelling rubber boats in fresh water and salt water service.

* 6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification

(b) Level of preservation, packaging and packing required (see 5.1 and 5.2)

* 6.2.2 Contract data requirements. Data conforming to Data Item Descriptions A-12, C-01, C-03, C-04, E-04, E-13, E-14, H-01, H-06, H-09, V-11 and V-16 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

* 6.3 Lubricants. When requested by the contractor, a list of available sources of standard lubricants should be provided by the contracting officer (see 3.3.12 and 3.3.13).

6.4 Color chips. Color chips for Marine Corps green, No. 23, may be obtained from Commanding General, Marine Corps Supply Activity (Code 843D), Philadelphia, Pennsylvania 19146.

* 6.5 First article instructions. The contracting officer shall specify the date and location to which the first article is to be delivered for examination and test. Records of the inspection for each unit shall be kept complete by the supplier and be made available to the Government upon request.

* 6.6 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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